

LAB EXPERIMENT RECORD

SET 5: Website Analytics

Department	AI&DS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	01 OCTOBER 2026
Day	THURSDAY

SET5:WebsiteAnalytics

Aim: To visualize website traffic data using a line chart, bar chart, and stacked area chart in R, and to build an interactive dashboard using Tableau.

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

PROGRAM 1: Line Chart (Daily Page Views)

Aim: To show the trend in daily page views using a line chart. `graphics.off()`

```
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
```

```
"2023-01-04", "2023-01-05"))
```

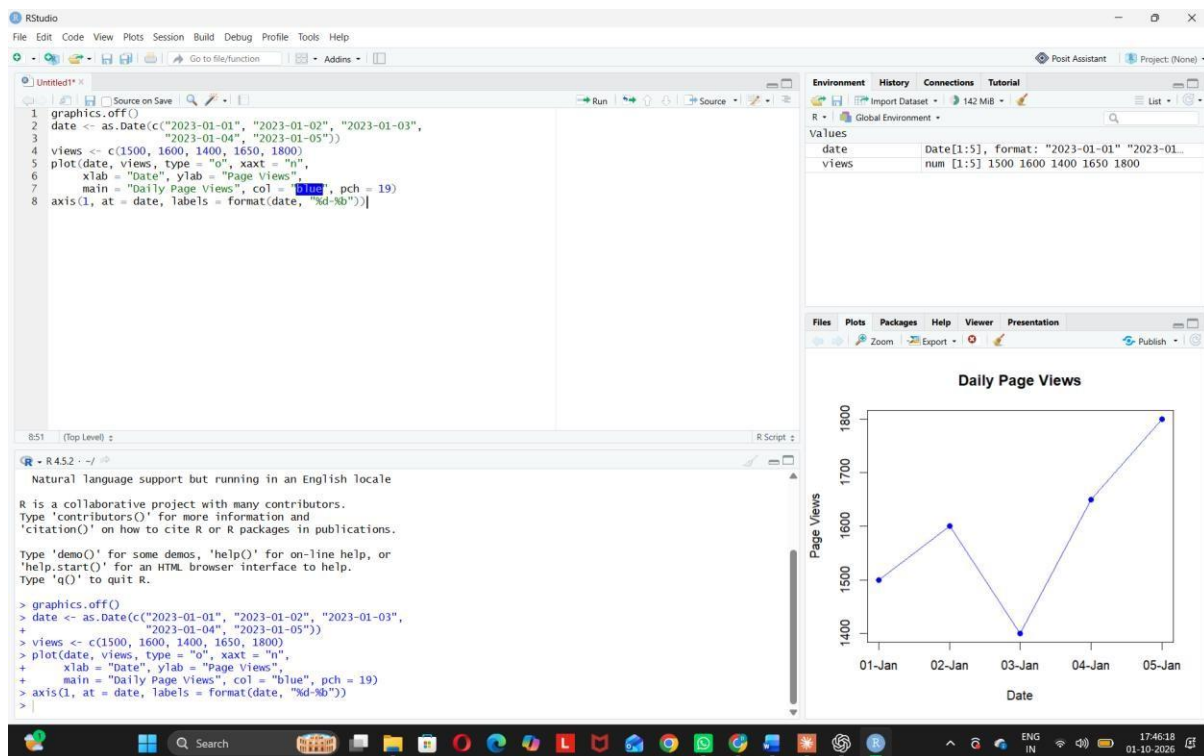
```
views<-c(1500,1600,1400,1650,1800)
```

```
plot(date,views,type="o",xaxt="n",xlab="Date",ylab=
```

```
"Page Views",
```

```
main="Daily Page Views",col="blue",pch=19) axis(1,at=date,labels=
```

```
format(date,"%d-%b"))
```



Observation: Page views dip on 3rd Jan and reach the highest (1800) on 5th Jan.

PROGRAM 2: Bar Chart (Top 3 Days by Click-through Rate)

Aim: To show the top N days with the highest click-through rates. `graphics.off()`

```
date<-c("01-Jan","02-Jan","03-Jan","04-Jan","05-Jan")
```

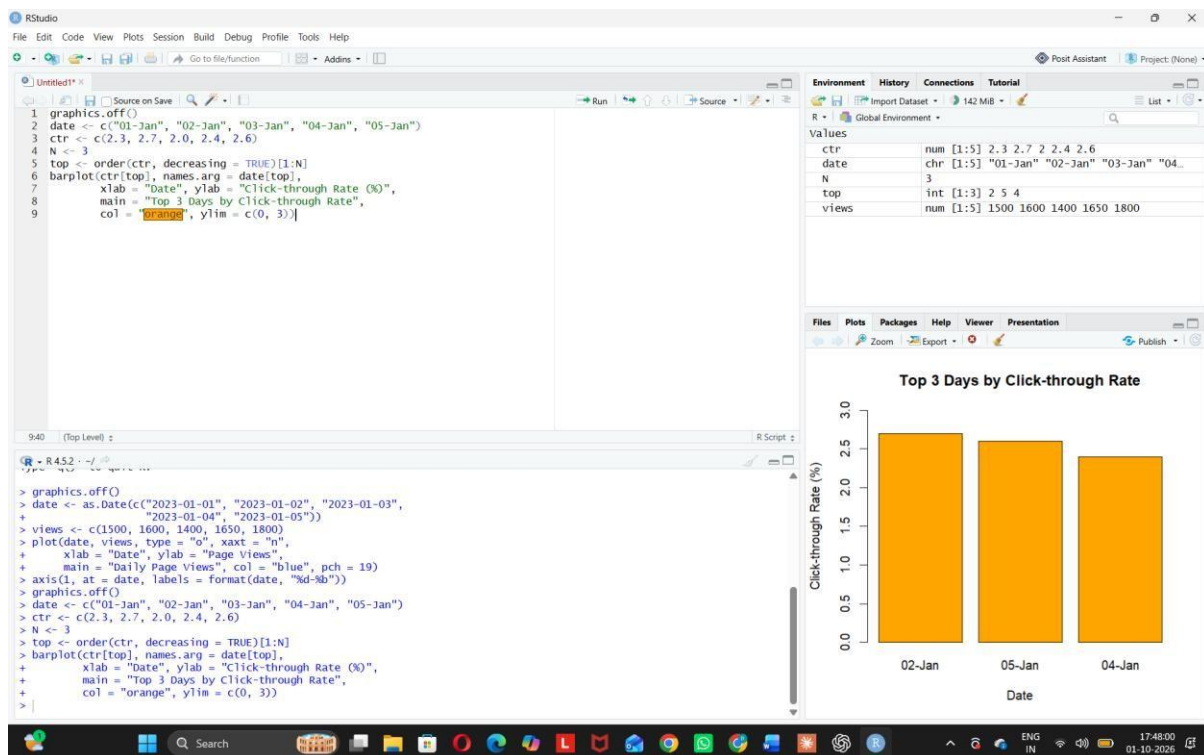
```
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)
```

```
N <- 3
```

```
top <- order(ctr, decreasing = TRUE)[1:N]
```

```
barplot(ctr[top], names.arg = date[top],
```

```
  xlab = "Date", ylab = "Click-through Rate (%)", main = "Top 3 Days by  
Click-through Rate", col = "orange", ylim = c(0, 3))
```



Observation: 2nd Jan has the best click-through rate. 3rd Jan (2.0%) is the lowest, so it is not in the top 3.

PROGRAM 3: Stacked Area Chart (User Interactions)

Aim: To show the distribution of likes, shares, and comments over time.

```
install.packages("ggplot2") #run only once library(ggplot2)

graphics.off()

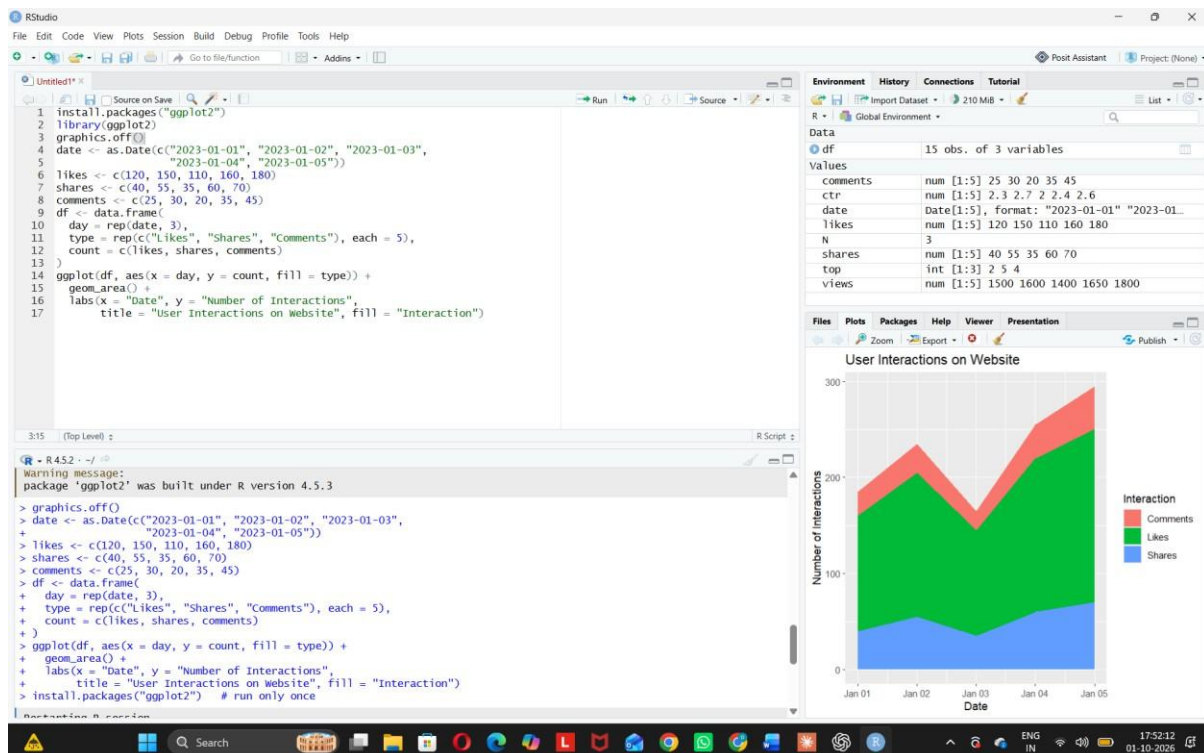
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                  "2023-01-04", "2023-01-05"))

likes<-c(120,150,110,160,180)

shares<-c(40,55,35,60,70)

comments<-c(25,30,20,35,45) df<-data.frame(
  day = rep(date, 3),
  type=rep(c("Likes", "Shares", "Comments"), each = 5), count = c(likes, shares,
  comments)
)

ggplot(df, aes(x = day, y = count, fill = type)) + geom_area() +
  labs(x = "Date", y = "Number of Interactions",
       title = "User Interactions on Website", fill = "Interaction")
```



Observation: Interactions are highest on 5th Jan. Most users like, fewer share, and the fewest comment.

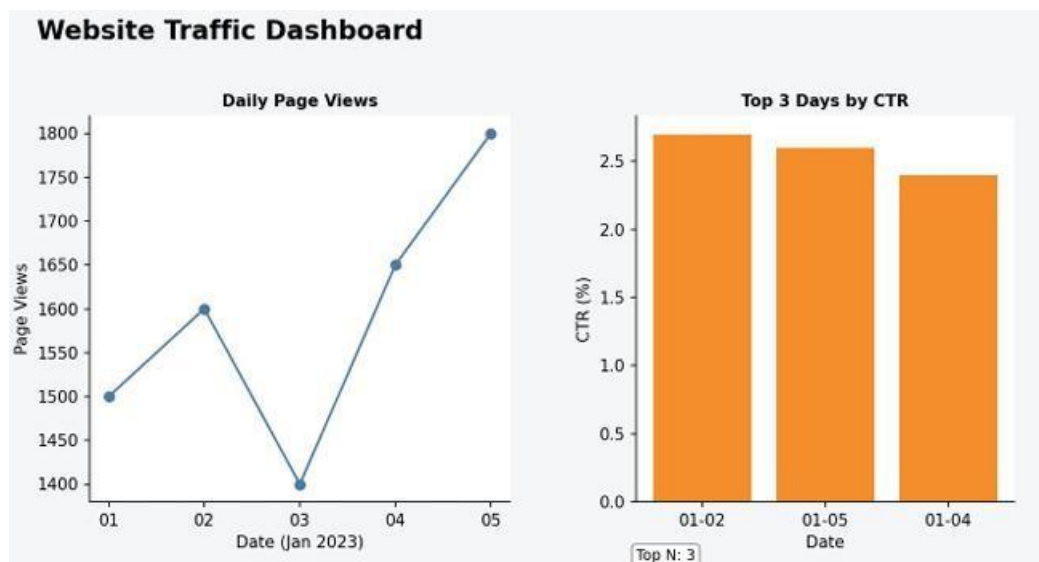
PROGRAM 4: Tableau Dashboard

Aim: To build an interactive dashboard with the line chart and bar chart in Tableau.

Steps in Tableau

1. Save the website traffic data (Date, Page Views, Click-through Rate) as an Excel or CSV file.
2. Open Tableau and connect to the file using Connect > Microsoft Excel/Textfile. Make sure Date has the Date data type.
3. Sheet 1 (Line Chart): Drag Date to Columns (set to Day, continuous) and Page Views to Rows (SUM). Add PageViews to Label and rename the sheet "Daily Page Views".
4. Create a parameter named "Top N" (Integer, value 3, range 1 to 5) and show the parameter control.

5. .Sheet 2(Bar Chart): Drag Date (Day, discrete) to Columns and Click-through Rate to Rows (AVG). Rightclick Date > Filter>Top, choose "Byfield", and set the value to the Top N parameter by Click-through Rate. Sort the bars in descending order and rename the sheet "Top N Days by CTR".
6. Create a new Dashboard (Dashboard>New Dashboard). Set the size to Automatic and drag both sheets onto the dashboard area.
7. Add interactivity: select each chart and click Use as Filter, so selecting a date in one chart filters the other chart. Add a Date range filter (Show Filter> Apply to Worksheets > Selected Worksheets).
8. Add the title "Website Traffic Dashboard", keep the Top N control visible, and test the dashboard in Presentation Mode.



Observation:

The dashboard shows the page view trend and the top days by click-through rate side by side. Users can change the Top N

value, select a date range, or click a point or bar to see how page views and click-through rate relate for a particular day.

Procedure

1. Define the date, page views and click-through rate vectors in R, converting the dates with `as.Date()`.
2. Use the `plot()` function with `type="o"` to create the line chart of daily page views, and `axis()` to label the dates.
3. Sort the data with `order()` and select the top N rows using `head()`, then use `barplot()` to create the bar chart.
4. Create a long-format data frame of likes, shares and comments, and use `ggplot2` with `geom_area()` to draw the stacked area chart.
5. Load the website traffic file into Tableau and build the line chart and top N bar chart worksheets.
6. Combine both worksheets in a Tableau dashboard and add the Top N parameter, filters and actions for interactivity.

RESULT:

The website traffic data was successfully visualized using a line chart, a bar chart and a stacked area chart in R. An interactive Tableau dashboard was also created, combining the line chart and the top N bar chart so that users can explore page views and click-through rates.